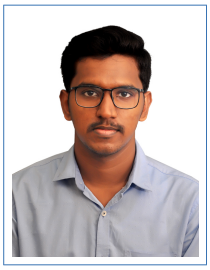
ARUN AMAITHI RAJAN

Curriculum Vitae

Senior Engineer
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Education

2022–2026 : **Doctor of Philosophy, Computer Science**, *College of Engineering, Guindy*, Anna University, Chennai.
Thesis: Towards Secure and Efficient Framework for Medical Image Encryption and Deep Hashing Based Retrieval in the Cloud

2018–2020 : **Master of Computing, Computer Science**, *School of Computing*, National University of Singapore, Singapore.

2014–2018 : **Bachelor of Engineering, Computer Science & Engineering**, *College of Engineering, Guindy*, Anna University, Chennai.

Publications - Journal Articles

2026 Aruna Gladys. A, Vetriselvi. V, Baskar P.N, and **Arun Amaithi Rajan**. Addressing Contextual Sparsity in Multimodal Emotion Recognition Using Speaker-Focused LLM Contextualization and Emotion-Driven Augmentation. *Array*, volume 30, page 100788. Elsevier, **IF: 5.3, Q1**.

2026 Aruna Gladys. A, Aishwarya. R, Vetriselvi. V, Daphne Samuel, Baskar P.N, and **Arun Amaithi Rajan**. Building Expert Small Models: A Comprehensive Survey of Model Compression, Knowledge Distillation, and Augmented Inference. *Archives of Computational Methods in Engineering*, volume 33, pages 10085–10126. Springer, **IF: 12.9, Q1**.

2025 **Arun Amaithi Rajan**, Vetriselvi. V, Mayank Raikwar, and Mohamed Fuzail. H. ESecMedIR: Efficient and Secure Dual-level Transformer based Medical Image Retrieval Framework in the Cloud. *Computers and Electrical Engineering*, volume 126, page 110519. Elsevier, **IF: 5.5, Q1**.

2025 **Arun Amaithi Rajan**, Vetriselvi. V, Ajitesh. M, and Praveen Kumar. R. Secure and Fault Tolerant Cloud Based Framework for Medical Image Storage and Retrieval in a Distributed Environment. *Scientific Reports*, volume 15. Springer, **IF: 4.9, Q1**.

2025 Praveen Kumar. R, Gautham Kumar. G, **Arun Amaithi Rajan**, Vetriselvi. V, and Mohamed Ihsan. P.M. A Comprehensive Security Framework for Cloud-Based Remote Sensing Image Storage and Retrieval With Adversarial Attack Resistance. *International Journal of Computers and Applications*, volume 47, pages 766–790. Taylor & Francis, **IF: 6.1, Q2**.

2025 Gautham Kumar. G, Praveen Kumar. R, **Arun Amaithi Rajan**, Vetriselvi. V, and Mohamed Ihsan. P.M. Healthchain: Protecting Healthcare Data Through Blockchain With Zero-Knowledge Proof and Biometric-Based Access Control. *Journal of Cyber Security Technology*, volume 10, pages 1–26. Taylor & Francis, **IF: 6.6, Q2**.

2024 **Arun Amaithi Rajan**, Vetriselvi. V, Mayank Raikwar, and Reshma Balaraman. SMedIR: Secure Medical Image Retrieval Framework With Convnext-Based Indexing and Searchable Encryption in the Cloud. *Journal of Cloud Computing*, volume 13. Springer, **IF: 5.5, Q1**.

2024 Ajitesh. M, Deekshith. M, **Arun Amaithi Rajan**, Vetriselvi. V, and Hemanth. D. EdgeShield: Attack Resistant Secure and Privacy-Aware Remote Sensing Image Retrieval System for Military and Geological Applications Using Edge Computing. *Earth Science Informatics*, volume 17, pages 2275–2302. Springer, **IF: 4.2, Q1**.

2023 **Arun Amaithi Rajan** and Vetriselvi. V. Systematic Survey: Secure and Privacy-Preserving Big Data Analytics in Cloud. *Journal of Computer Information Systems*, volume 64, pages 136–156. Taylor & Francis, **IF: 6.6, Q1**.

Publications - In Conference Proceedings

2026 Sudeep Ravichandran, Riya Elizabeth Kuriyan, Aruna Gladys. A, Vetriselvi. V, and **Arun Amaithi Rajan**. LogREx: Reasoning-Enabled LLM and Knowledge Graph Based Anomaly Detection in System Logs. In *Sixth Congress on Intelligent Systems*, pages 24–37. Springer.

2025 **Arun Amaithi Rajan** and Vetriselvi V. Privacy or Performance? Towards Secure and Scalable Medical Image Storage and Retrieval Management in the Cloud. In *Proceedings of the 18th Innovations in Software Engineering Conference*, pages 1–4. ACM.

2025 Riya Elizabeth Kuriyan, Sudeep Ravichandran, **Arun Amaithi Rajan**, and Vetriselvi V. PCMedIR: Privacy-Enhancing Cross-Modal Medical Information Retrieval System in Cloud. In *Communication and Intelligent Systems*, pages 361–376. Springer.

2025 Anand. B, **Arun Amaithi Rajan**, and Vetriselvi V. A^2RS: Adversarial Attack Resistant Steganography Technique for Medical Images in Digital Healthcare. In *Computational Intelligence in Data Science*, pages 339–353. Springer.

2024 **Arun Amaithi Rajan**, Vetriselvi. V, Aishwarya. R, and Anitha Amaithi Rajan. Efficient and Lightweight Framework for Confidential Medical Image Search using Edge Computing. In *2024 International Conference on Computational Intelligence and Network Systems (CINS)*, pages 1–7. IEEE.

2024 Mohamed Ihsan. P.M, **Arun Amaithi Rajan**, Vetriselvi. V, Gautham Kumar. G, and Praveen Kumar R. QCrypt: Advanced Quantum-based Image Encryption for Secure Satellite Data Transmission. In *2024 Third International Conference on Electrical, Electronics, Information and Communication Technologies (ICEEICT)*, pages 1–9. IEEE.

2024 Barvin. D.S, Anitha Amaithi Rajan, and **Arun Amaithi Rajan**. WanderGenie: A Smart Guide for Immersive Tourism through Generative AI. In *2024 International Conference on Computational Intelligence and Network Systems (CINS)*, pages 1–6. IEEE.

2023 **Arun Amaithi Rajan**, Vetriselvi. V, and Aruna Gladys A. Secure Image Encryption Model for Cloud-Based Healthcare Storage Using Hyperchaos and DNA Encoding. In *Computational Intelligence in Data Science*, pages 89–103. Springer.

2019 Vetriselvi. V, Pragatheeswaran. S, Thirunavukkarasu. V, and **Arun Amaithi Rajan**. Preventing Forgeries by Securing Healthcare Data Using Blockchain Technology. In *Information and Communication Technology for Sustainable Development*, pages 151–159. Springer.

Research Experience

College of Engineering Guindy, Anna University, Chennai

Jan, 2022 – May, 2025 ***Secure and Privacy-Preserving Medical Image Retrieval in Cloud Environments***.
Developed deep learning and cryptography-based frameworks for secure medical image encryption and content-based retrieval in cloud environments. Integrated selective encryption, deep hashing, and hybrid mechanisms to achieve privacy-preserving, accurate, and efficient medical image retrieval.

Advisor : **Dr. Vetriselvi V**, *Professor, Department of Computer Science & Engineering*, College of Engineering Guindy, Anna University (*Personal Web-page*)

Paypal, Singapore

May, 2019 – Sep, 2019 ***Hardware-Assisted Confidential Computing Using Intel SGX***.
Designed and developed a PoC leveraging Intel Software Guard Extensions (SGX) to explore hardware-based trusted execution environments (TEEs) for secure and isolated computation in cloud environments.

Advisor : **Dr. Hubert Le Van Gong**, *Managing Director, Cybersecurity Division*, JPMorgan Chase and Co (*Linkedin Page*)

Industry Experience

Qualcomm Technologies, Chennai

June, 2025 – present — ***Senior Software Engineer - Hypervisor Team***.
Working on hypervisor development and hardware-assisted security, focusing on Hafnium, ARM Secure EL2, FF-A, and ARM CCA. Contributing to secure system design, CI/CD automation, and technical mentoring of junior engineers.

Mircon Technologies, Singapore

Jan, 2020 – Jan 2022 — ***Firmware Development Engineer - SSD Security Team*** .
Developed security firmware for SSD platforms, focusing on Secure Boot, hardware attestation, eFuse-based security, firmware authentication, and revocation mechanisms. Worked on security standards and technologies including TCG and FIPS.

Awards & Recognitions

2026 Received the ***Go-Getters Award*** in recognition of outstanding contributions and successful delivery of key initiatives within the Hypervisor team.

2024 Received the ***Best Paper Award*** for the QCrypt research article at ICEEICT 2024, Tiruchirappalli.

2023 Received the ***Best Paper Award*** for the hyperchaos-based image encryption research article at ICCIDS 2023, Chennai.

2021 Received the ***Employee Achievement STAR Award*** in recognition of outstanding contributions to the SSD Security team and successful delivery of key security initiatives for the Data Centre project.

Invited Talks

Apr, 2026 *Secure IoT Systems and Firmware Security: Bridging Artificial Intelligence and Cybersecurity* at VIT, Chennai

Mar, 2026 *Secure Healthcare Data Processing with Confidential Computing* at VelTech, Chennai

Dec, 2025 *Confidential Computing* at College of Engineering Guindy, Anna University, Chennai

Nov, 2024 *EdgeShield: Secure Edge Computing* at International Conference on Space Science and Technology, Dubai

Technical skills

Languages: C, C++, Embedded C, Python, Go, Java

Frameworks: PyTorch, TensorFlow, Scikit-learn, Open SSL

FW and Security: NVMe, TCG, Secure boot and download, Key Management

Teaching Assistantship

Fall, 2023 : **CS6104: Data Structures and Algorithms**, CEG.

Fall, 2018 : **TIC1001: Introduction to Computing and Programming I**, NUS.

Fall, 2017 : **GE8151: Computing Techniques**, CEG.

Project Mentorship

Spring, 2025 : **Blockchain-Based Authenticated Image Storage with Steganography**, CEG, Postgraduate.

Spring, 2025 : **LLM-Driven Cybersecurity and Intelligent Anomaly Detection**, CEG, Undergraduate.

Spring, 2024 : **Adversarial Attack Resistant Storage and Retrieval**, CEG, Undergraduate.

Spring, 2023 : **Privacy-aware Image Storage and Retrieval**, CEG, Undergraduate.

Referees

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Dr. J. Angel Arul Jothi
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BITS Pilani
Dubai Campus
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